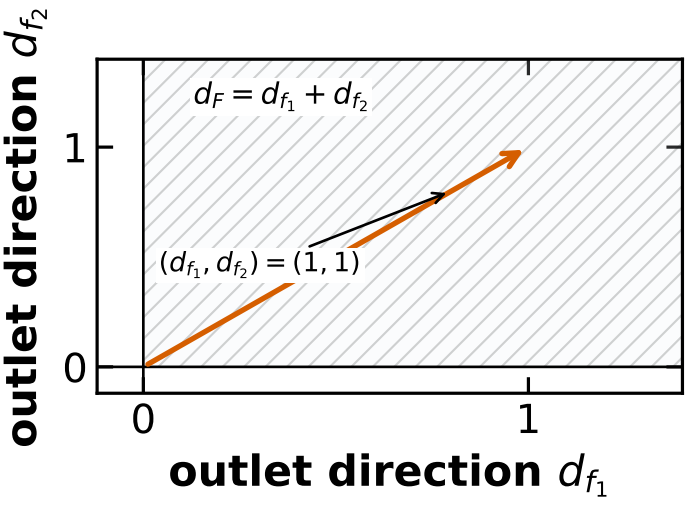

At zero flow, the active bound rows are

$$\mathbf{r}_1 = (-1, 0, 0)$$

$$\mathbf{r}_2 = (0, -1, 0)$$

$$\mathbf{r}_3 = (0, 0, -1)$$

The balance row is $(1, -1, -1)$.